Adult Income Baseline: Summary Report

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1. Problem Framing

Target: Predict whether a person earns more than 50K a year (positive class).

Business use case: A targeted marketing campaign for a premium product, where every contact costs money, so wasted outreach to the wrong person is a real cost, not just a missed opportunity.

Class base rate: 23.9% of people in the dataset earn more than 50K. This means a model that always predicted “no” would still be 76% accurate without learning anything, which is why accuracy alone is not used as the scoring metric.

Primary metric: Precision. Every false positive is a wasted contact, so precision (of the people flagged as high earners, what fraction actually are) directly measures the cost the business cares about most, and is the number used to compare baselines and models across the week.

2. Reproducible Splits

Data was split into train (34,188 rows), dev (4,885 rows), and a hold-out test set (9,769 rows, ~20% of the data), stratified on the target with a fixed random_state=42. The hold-out test set is only used for final evaluation. Using it during tuning would let our own tuning decisions leak into the test score, turning it from an honest estimate of real-world performance into an overly optimistic one.

3. Baseline Results (on hold-out test set)

Baseline	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority class	0.761	0.000	0.000	0.000	-	-
Rule: edu-num>=13 OR cap-gain>0	0.745	0.475	0.603	0.531	0.697	0.381

Three candidate rules were compared before picking a baseline: capital-gain > 0 (precision 0.62, recall 0.21), education-num >= 13 (precision 0.48, recall 0.50), and the combined rule education-num >= 13 OR capital-gain > 0 (precision 0.48, recall 0.59). The combined rule was chosen for a noticeably higher recall while precision stayed roughly level with education alone. The majority-class baseline scores 0.0 on precision, recall, and F1 despite 76.1% accuracy, it never predicts a positive so it is not usable for the business goal. Any real model built this week needs to beat the combined rule's precision (0.475) by a meaningful margin to be worth deploying.

4. Error Analysis

On the hold-out test set the rule produced 1,561 false positives and 928 false negatives. False positives skew slightly younger (mean age 38.7 vs 44.0 for false negatives) and average the same 40 hours per week. False negatives skew older, work slightly longer hours (mean 44.5), and are overwhelmingly married (91.5% are Married-civ-spouse), suggesting marital-status carries signal the rule cannot see. Most false positives are degree holders or capital-gain earners in lower-paying occupations or early-career roles, where a credential or a one-off investment gain does not yet translate into income.

Data and feature issues to fix before modeling:

- workclass, occupation, and native-country missing values were filled with an explicit ‘Unknown’ category; worth testing whether model-based imputation performs better
- education and education-num are redundant, keep one
- capital-gain and capital-loss are heavily skewed (mostly zero), consider log-transform or bucketing
- fnlwgt is a census sampling weight with no individual-level meaning, likely drop it
- native-country is high cardinality dominated by the US, needs grouping rather than raw one-hot encoding
- all categorical features need proper encoding before any model beyond a simple rule can use them

Metric to optimize for the rest of the week:

Precision. The combined rule baseline sets the bar at precision = 0.475. Every model built this week should be compared against this number using the same hold-out test set, held back until final evaluation.